

ARCADE: Adaptive Reasoned Composition for Anomaly Detection in Graphs

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Abstract—No single unsupervised method detects anomalies well on every graph: each detector assumes one anomaly mechanism, and every assumption fails somewhere. Since there are no labels, choosing the right detectors for a given graph is a reasoning task, usually done by a human expert. ARCADE automates this role. It combines a library of detectors that covers the main anomaly mechanisms, a label-free *evidence interface* that describes the graph to an analyst, a language-model analyst that selects a small set of detectors from this evidence, and a label-free gate that composes their scores. We state four constraints such a system must satisfy (coverage, diagnosability, complementarity, and fusability) and turn each into concrete design rules, validated by ablations of the final system. On 10 benchmarks ARCADE reaches a mean AUROC of 81.8 and exceeds every self-run baseline on 8 of the 10 benchmarks. A study of 10 language models in the analyst seat shows that success depends less on model scale than on the evidence they are given: with the full evidence interface, an 8-billion-parameter local model identifies the correct mechanism on all 8 test graphs and surpasses running the entire library, at a fraction of the compute. Code: <https://anonymous.4open.science/r/arcade>.

Index Terms—graph anomaly detection, large language models, unsupervised learning, detector composition, graph neural networks

I. INTRODUCTION

Detecting anomalous nodes in attributed graphs matters for fraud detection, intrusion detection, and social network analysis [1]. Many unsupervised detectors exist, from reconstruction autoencoders [2], [3] to contrastive [4], [5], one-class [6], and neighborhood-based methods [7]. Benchmarks agree on one point: none of them wins everywhere [8], [9]. Each detector encodes one assumption about what an anomaly is, and each assumption holds on some graphs and fails on others. Our own measurements confirm this: across 10 benchmarks, the best baseline changes from dataset to dataset, and every baseline collapses on at least one (Table II).

In the unsupervised setting this creates a problem that is not about architectures. With no labels, someone has to decide which detectors fit the graph at hand. In practice this is a human expert who inspects the graph and reasons about its likely anomaly mechanisms. ARCADE (Adaptive Reasoned Composition for Anomaly Detection) automates this role. A language-model analyst receives a label-free description of the graph: structural and attribute statistics, the catalog of applicable detectors with the evidence for each, and score distributions of cheap detectors. It reasons about which mechanisms this evidence supports and selects a small set of detectors. A label-free gate then composes their scores. Labels are used exactly once, at the end, to measure the result.

Building such a system raises a design question that we found more important than any single component: *what must be true for reasoned composition to work at all?* We identify four constraints. The library must *cover* the mechanisms the analyst can meet; each mechanism must be *diagnosable* from label-free evidence; fused detectors must be *complementary*; and the composition rule must be *fusable*, meaning it combines scores without diluting a strong signal or trusting a degenerate one. Each constraint translates into design rules, and each rule is validated by an ablation of the final system, including a counterintuitive one: adding a quantitative “quality” score to the evidence makes analysts worse, because they rank by the number instead of reasoning about the mechanism.

Our contributions are:

- *A framework for reasoned composition.* A library of 16 detection paradigms (11 fast, 5 deep) plus a component grid of encoders, objectives, decoders, and augmentations; a label-free evidence interface; an exchangeable language-model analyst; and a label-free composition gate. The whole system is exposed as a tool server that any agent can operate.
- *Design constraints with validated rules.* Coverage, diagnosability, complementarity, and fusability, each analyzed by controlled ablations: removing one evidence field makes all analysts fail on the affected graph; adding a quality score unfixes previously correct decisions; replacing the union rule by an average costs 12 AUROC points on one benchmark.
- *A study of the analyst seat.* Ten language models, from a 2B local model to frontier models, under identical evidence and protocol, with every prompt, decision, and rationale logged. Success tracks deliberate evidence reading, not benchmark rank: with full evidence, qwen3:8b (a free 5 GB local model) identifies the correct mechanism on 8 of 8 graphs and outperforms running the whole library, while the strongest general-purpose model misses the two graphs whose answers contradict standard anomaly priors.
- *Honest system results.* On 10 benchmarks, mean AUROC 81.8 with zero seed variance on half of them; above every self-run baseline on 8 of 10 benchmarks; the main disclosed failure (Amazon) is largely recovered by the analyst mode.

II. RELATED WORK

A. Unsupervised graph anomaly detection

Reconstruction methods such as DOMINANT [2] and AnomalyDAE [3] score nodes by how poorly an autoencoder reconstructs them. Contrastive methods such as CoLA [4] and SL-GAD [5] compare a node with its sampled local

subgraph. One-class methods adapt Deep SVDD to graphs [6], [10]; DONE [11] and GAAN [12] use dedicated autoencoding and adversarial objectives; GAD-NR [7] reconstructs neighborhoods. Shallow methods stay competitive in their regime: Radar [13] uses a structured linear residual, TAM [14] scores local affinity. Surveys [1], [15] and benchmarks [8], [9] document that each family has graphs where it fails, which is the premise of this work.

B. Automated anomaly detection

AutoGAD [16] uses a language model to search the configuration space of one reconstruction detector. MetaOD [17] selects one detector by meta-learning on tabular data. AD-AGENT [18] and Court-of-LLMs [19] orchestrate tabular and text pipelines. These systems search or select a single method. ARCADE differs in two ways: it composes several complementary detectors, and it treats the evidence given to the model as a designed, measurable part of the system. Language models on graphs are surveyed in [20].

C. Ensemble anomaly detection

Score-level ensembles are well studied on tabular data: SELECT [21] prunes detectors that disagree with the consensus, AOM [22] averages group maxima, UED [23] weights by tail agreement, and score unification [24] makes outputs comparable. ARCADE’s gate builds on these ideas and adds label-free rules for when to select, union, or average, driven by the constraint analysis of Section IV.

III. THE ARCADE FRAMEWORK

ARCADE has four parts (Fig. 1). A *detector library* covers the main anomaly mechanisms. An *evidence interface* describes the loaded graph without labels. An *analyst* (any language model) reads the evidence and selects detectors. A *gate* composes the selected scores, again without labels. Ground-truth labels appear only when we evaluate the final ranking.

A. Detector library

The library provides 16 detection paradigms (Table I). Eleven are fast (seconds): local affinity [14], density, community deviation, homophily violation, spectral response, attribute–structure mismatch, and 5 closed-form detectors that compute a fixed formula on the graph (conformity via truncated-SVD residual, a Radar-style linear residual [13], peripheral mismatch for very small graphs, attribute inflation for low-dimensional count features, and clique density for dense substructures). Five are deep (minutes on one GPU): reconstruction [2], contrastive [4], adversarial [12], one-class [6], and subgraph-contrastive in the CoLA/SL-GAD family [5]. Our subgraph-contrastive implementation matches its reference under an identical protocol (AUROC 88.4 vs. 88.7 on the Cora benchmark, same seed and budget).

Every paradigm carries a *precondition*: a label-free predicate over the graph profile that states when its assumption is meaningful. For example, the linear residual requires dense continuous features (sparsity ≤ 0.3 , dimension ≥ 100); conformity

TABLE I: The detector library: every paradigm with its type (C = closed-form formula, F = fast non-parametric, D = deep/trained), the assumption it scores by, and the label-free precondition under which it runs.

Paradigm	T	Anomalies are nodes that...	Precondition
reconstruction	D	are hard to reconstruct	$\dim \geq 2$
contrastive	D	disagree across views	$n \geq 50$
adversarial	D	look generated to a critic	$n \geq 500$
one-class	D	fall outside the normal ball	$\dim \geq 2$
subgraph contrast	D	misfit their own local sub-graph	$\dim \geq 100$
local affinity	F	disagree with their neighbors	$\text{hom.} \geq 0.3$
density	F	sit in low-density regions	$n \geq 30$
community	F	fit no community well	≥ 2 comms
homophily violation	F	mismatch their neighborhood	$\text{hom.} \geq 0.4$
spectral	F	carry high-frequency energy	$n \geq 30$
attr–struct mismatch	F	rank differently by attr. vs. struct.	$\dim \geq 2$
conformity	C	are the most template-like rows	$\text{dup} \geq 0.4$
linear residual	C	resist a structured linear model	$\text{spars} \leq 0.3, \dim \geq 100$
peripheral mismatch	C	combine small-graph periphery cues	$n \leq 500$
attribute inflation	C	are high on many attributes	$\text{deg} \leq 5, \dim \leq 100$
clique density	C	sit in abnormally dense subgraphs	$\text{spars} \geq 0.9, \dim \geq 100$

requires a high duplicate-row rate (≥ 0.4); clique density requires sparse high-dimensional features. Preconditions guard against running a detector in a regime where its output is noise. Beyond the paradigms, trained methods can be assembled from a component grid of 5 encoders (GCN [25], GIN [26], GAT [27], MLP, and a community-aware encoder in the style of [28]), 10 objectives, 4 decoders, and 6 augmentations.

B. Evidence interface

The analyst never sees the graph itself, and never sees labels. It sees an *evidence pack* with three parts.

(1) *Graph profile*. Structural statistics (size, density, average and maximum degree, degree spread, clustering), community statistics (Louvain [29] count, modularity), attribute statistics (dimension, sparsity, and the duplicate-row rate), and homophily. The rule for this list is simple: *every quantity that any precondition reads must be present*. A single missing field can make a mechanism unreachable for every analyst (Section VI-C).

(2) *Grounded catalog*. The applicable paradigms, each with its assumption in one sentence and an *applicability grounding* that instantiates the precondition with the actual values, for example: “applicable because $\text{feature_dup_rate} = 0.567$ satisfies ≥ 0.4 ”. Stating why a precondition holds is evidence, not advice; the selection among the grounded candidates stays with the analyst.

(3) *Landmarkers*. For each fast paradigm, the quantiles of its score distribution on this graph. A strongly separated upper tail suggests that the paradigm’s mechanism is present. The pack deliberately reports *only raw quantiles*, not any derived quality score: a per-paradigm quality number acts as a false ranking signal that displaces mechanism reasoning (Section VI-C).

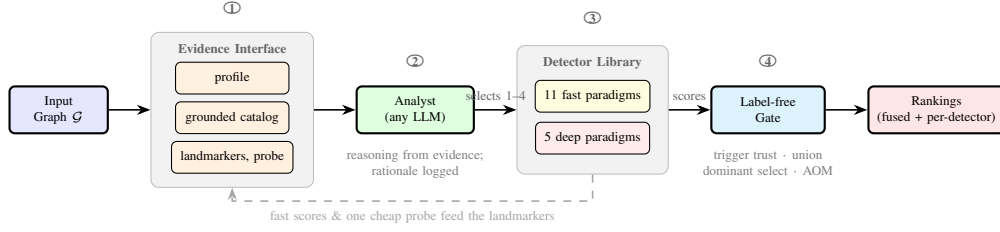


Fig. 1: The ARCADE workflow. ① The evidence interface describes the graph without labels: the profile, the applicable-paradigm catalog grounded with the actual values, and score-distribution landmarks (the fast detectors and a cheap probe of the deep subgraph-contrastive view). ② The analyst (any language model) reasons from this evidence and selects 1–4 paradigms, logging its rationale. ③ The selected detectors run. ④ The label-free gate composes their scores and outputs both the fused ranking and every detector’s own ranking. Labels are used once, after this pipeline, for evaluation only.

Deep paradigms would otherwise be invisible at selection time, so each applicable deep paradigm contributes a *probe*: a very short run (5 epochs, 4 test rounds), clearly labeled as a cheap preview, whose quantiles enter the pack.

The pack contains no dataset names and nothing about how the benchmark was constructed. The analyst cannot pattern-match benchmark folklore; it can only reason from the evidence.

C. Analyst

The analyst is any language model. It receives the evidence pack and must return, as structured output, a selection of 1–4 paradigms and a written rationale. The rationale is stored with the decision; every claim about analyst behavior in this paper is backed by these stored traces. Two protocols are supported. In the *free-form* protocol the model selects directly. In the *checklist* protocol it must first give a verdict (select or reject, with evidence) for every applicable paradigm, then select; this converts open recall into verification, which weaker models handle better (Section VI-D). The analyst is exchangeable: Section VI-B evaluates 10 models from a 2B local model to frontier models under identical evidence.

D. Composition gate

The gate receives the scores of the selected paradigms and outputs one ranking, without labels. Scores are first normalized per paradigm. The gate then applies four rules, in order.

Trigger trust. A closed-form detector runs only when its precondition holds, and the precondition holding is itself label-free evidence that its mechanism is present. If a triggered closed-form detector is among the selection, the gate trusts it. When the trigger is clique density, a structural mechanism that often coexists with a contextual anomaly population, the gate pairs it with a contextual companion chosen by degree regime (the subgraph-contrastive view on dense graphs, local affinity on sparse ones). The companion must pass two label-free checks: its score must not simply echo node degree (rank correlation with degree < 0.6), and it must have a separated tail of its own. A companion that fails contributes no retrievable population to a union, only noise; if no candidate passes, the gate abandons the trusted pair and falls back to the general rules below.

Union for two populations. A trusted structural+contextual pair is fused by the per-node *maximum* of the two rank scores,

not their average. The two views retrieve different anomaly populations; the union keeps both, the average dilutes both. This single rule is worth 12.1 AUROC points on ACM (98.9 vs. 86.8) and 2.4 on Cora (Section VI-F).

Dominant selection. If the same paradigm leads both label-free quality diagnostics (tail separation and UED tail quality [23]) and its UED is at least twice the runner-up’s, the gate uses it alone. Averaging a dominant signal with weaker correlated ones can only blur it (93.7 vs. 88.9 on the Amazon pair). Requiring the conjunction of two diagnostics matters, because either one alone can mislead.

Diverse ensemble. Otherwise, if the selected paradigms are diverse (low mean pairwise rank correlation) with no dominant member, the gate uses average-of-maxima [22] over canonically ordered groups; weighted averaging is reserved for the case where trained components clearly dominate. All gate computations are deterministic and invariant to the order in which scores arrive.

E. Tool server

The entire workflow is exposed as a Model Context Protocol server with 66 tools (profiling, catalog, detector execution, diagnostics, gate, evaluation). Any agent, frontier or local, can operate the same workflow; our analyst study drives it programmatically through this interface.

IV. DESIGN CONSTRAINTS

An adaptive composition system fails in characteristic ways, and each failure traces to one of four constraints. We state each constraint, the design rules that enforce it, and the ablation that validates the rules (numbers in Section VI).

Coverage. For every anomaly mechanism the analyst may meet, the library must contain a detector whose assumption separates it. No selection policy can recover a mechanism outside the library. Coverage also has a quality dimension: a detector that under-performs its reference implementation silently shrinks coverage, so library detectors are validated against their references under identical protocols (Section III-A).

Diagnosability. Each covered mechanism needs label-free evidence that makes it reachable by reasoning. This constraint has two separable parts, and our evidence ablation (Section VI-C) isolates both: the *interface* must expose the relevant fields (with the duplicate-row rate hidden, zero of

eight analysts find the conformity mechanism; with it, four do), and the *analyst* must connect evidence to mechanism (models differ sharply on inferences that contradict standard anomaly priors). Diagnosability is also where evidence design can backfire: quantitative quality scores in the pack *reduce* correct decisions.

Complementarity. Fusing pays only when the fused detectors target different anomaly populations. The gate measures this by rank correlation and by the two-population structure of trusted pairs; the union rule exists exactly for the complementary case.

Fusability. Scores must be composable without diluting a dominant signal (dominant selection), without trusting a degenerate partner (companion checks), and reproducibly (deterministic, order-invariant fusion). Each rule corresponds to a measured failure of the corresponding naive choice.

On the numeric thresholds. Preconditions and gate rules use fixed constants. Each one encodes a qualitative regime rather than a tuned value: features that are mostly duplicated, mostly zero, or too few nodes to train on; a score that mostly follows degree. The constants are set once, globally, and never adjusted per dataset, and the decisions they produce are insensitive to their exact values: we verified that varying each constant across the gap between the two observed groups of profile values changes no decision on any benchmark. On a new graph that falls near a boundary, the consequence is bounded: the affected detector simply runs or does not, and the gate’s checks still apply downstream.

V. EXPERIMENTAL SETUP

Datasets. Seven standard benchmarks with ground-truth anomalies from PyGOD [30]: Enron, Reddit, Books, Disney, Weibo (organic anomalies), and Amazon and Cora (injected anomalies); plus three dense social benchmarks with standard injections, BlogCatalog, ACM, and Flickr [4]. All ten are treated identically; the analyst never learns a dataset’s name or provenance.

Baselines. DOMINANT, AnomalyDAE, DONE, CoLA, GAAN, and OCGNN, all run on our hardware with library defaults over 5 seeds [30]; SL-GAD from a faithful port of the official code (validated at 89.5 vs. its published 91.3 on its own Cora setup) over 5 seeds, using 128 evaluation rounds, where we verified on its two strongest datasets that the full 256-round reference budget gives identical results (Amazon 93.0 ± 1.1 vs. 92.9 ± 1.1 ; Cora 86.4 ± 1.4 in both); and AutoGAD [16] from its released code with a budget of 60 evaluations. AutoGAD selects configurations by measured AUROC, so it uses labels during search; we mark it accordingly and exclude it from best-baseline comparisons. On the highest-dimensional graphs, fitting is batched where full-batch training exceeds GPU memory; cells that remain infeasible on our hardware, or fail with runtime errors, are marked “/”.

ARCADE configuration. The system result runs every applicable paradigm and lets the gate compose (no language model involved); the analyst result uses a language model

to select 1–4 paradigms first. Deep detectors use fixed budgets (100 epochs; 256 evaluation rounds for the subgraph-contrastive paradigm) on one NVIDIA RTX A2000 (12 GB). The metric is AUROC, mean $\pm$ std over 5 seeds; closed-form detectors are deterministic, so several results carry zero seed variance by construction. Per-cell average precision (AP) and a Friedman–Nemenyi significance analysis over Table II are provided with the code.

Analyst models. Six local models via ollama: gemma4:e2b and gemma4:12b; qwen3:4b and qwen3:8b; llama3.1:8b; phi4:14b. One free hosted model: deepseek-v4-flash. Three Claude models via separate headless sessions: claude-haiku-4.5, claude-sonnet-5, claude-opus-4.8 (no extended reasoning; exact versions logged with the code). Local decisions are greedy (temperature 0) with 3 repeats; decision stability is measured separately at sampling temperature 0.7 (local) or the provider’s default sampling (hosted). Detector scores are computed once per benchmark with a fixed seed, and every arm’s selection (language model or baseline policy) is composed by replaying these cached scores through the same label-free gate, so arms differ only in what they select. Every prompt, response, rationale, latency, and cost is logged and shipped with the code.

VI. RESULTS

A. Detection quality

Table II reports all 10 benchmarks under one protocol, with per-method averages in the bottom row. ARCADE has the highest AUROC on 8 of 10 and the highest average (81.8). On 5 datasets the gate settles on closed-form detectors and the result carries zero seed variance. Amazon is one of the two losses: the injected anomalies there are best captured by a single strong subgraph-contrastive view, and running everything dilutes it; the label-free dominant-selection rule fires only for small selections, not for the full 11-paradigm pool. We report this as a limitation of the run-everything mode; the analyst mode recovers most of the loss (Section VI-B). On Flickr, the other loss, ARCADE is 1.7 below the strongest baseline. Two reproducibility notes on the three social graphs, whose published baselines come from the CoLA/SL-GAD literature. First, SL-GAD (self-run through a validated port of the official code) is stable across seeds but lands 1–4 points below its published numbers there. Second, PyGOD’s CoLA reimplementations omits the multi-round test-time averaging this method family requires, and its low column on these graphs reflects that omission, not the method: our own faithful implementation of the same family reaches 77.8–80.9 on the same graphs (the subgraph-contrast row of Table III), in line with CoLA’s published range. On ACM the union rule retrieves both injected populations almost perfectly (AP 80.5).

Table III shows every applicable detector on every benchmark, next to the composed result. Three observations. First, the table restates the paper’s premise inside the library itself: every column has a different winner, every row fails somewhere, and many cells sit *below* chance (local affinity at 25.7 on Weibo, contrastive at 38–44 on the injected graphs). A

TABLE II: AUROC (%) on all 10 benchmarks, mean $\pm$ std over 5 seeds, every method run on our hardware. AutoGAD[†] selects by measured AUROC (label-guided) and is excluded from the best-baseline comparison (underlined). “/” = runtime failure or infeasible on our hardware (12 GB GPU) even with batched fitting. A std of 0.0 means below 0.05 after rounding (seeds vary initialization and sampling). The Average row is the mean AUROC across benchmarks; * marks a mean over the graphs the method completed (DOMINANT 7, GAAN 8, AutoGAD 9 of 10), which if anything favours those methods by dropping their hardest cases.

Dataset	DOMINANT	AnomalyDAE	DONE	CoLA	GAAN	OCGNN	SL-GAD	AutoGAD [†]	ARCADE
Enron	55.5 $\pm$ 1.1	48.4 $\pm$ 11.6	41.1 $\pm$ 0.6	56.3 $\pm$ 3.3	49.2 $\pm$ 8.8	57.6 $\pm$ 1.8	53.1 $\pm$ 3.2	69.4	83.7$\pm$0.0
Reddit	56.1 $\pm$ 0.0	49.0 $\pm$ 1.3	53.3 $\pm$ 0.9	53.9 $\pm$ 2.3	49.3 $\pm$ 0.7	50.6 $\pm$ 8.4	53.7 $\pm$ 1.5	56.4	59.8$\pm$0.7
Books	47.7 $\pm$ 2.3	41.9 $\pm$ 1.1	37.6 $\pm$ 2.4	50.8 $\pm$ 5.8	50.3 $\pm$ 4.5	50.9 $\pm$ 0.4	<u>53.6$\pm$3.8</u>	63.9	72.1$\pm$0.0
Disney	59.7 $\pm$ 7.8	48.5 $\pm$ 0.1	40.8 $\pm$ 0.6	50.5 $\pm$ 13.1	48.0 $\pm$ 0.0	66.9 $\pm$ 0.6	58.9 $\pm$ 3.5	67.4	86.4$\pm$0.0
Weibo	90.0 $\pm$ 0.5	87.6 $\pm$ 1.1	85.2 $\pm$ 0.2	24.0 $\pm$ 2.1	89.9 $\pm$ 0.0	21.7 $\pm$ 2.6	49.9 $\pm$ 1.3	93.3	94.9$\pm$0.0
Cora	76.6 $\pm$ 0.4	71.4 $\pm$ 0.2	84.3 $\pm$ 0.8	55.0 $\pm$ 2.7	/	50.0 $\pm$ 0.0	86.4 $\pm$ 1.4	84.9	93.6$\pm$0.0
Amazon	71.8 $\pm$ 0.0	75.1 $\pm$ 0.0	84.8 $\pm$ 3.1	60.9 $\pm$ 1.4	/	50.1 $\pm$ 0.1	92.9$\pm$1.1	83.8	73.2 $\pm$ 0.1
BlogCatalog	/	74.7 $\pm$ 0.0	77.2 $\pm$ 0.7	49.5 $\pm$ 3.0	50.1 $\pm$ 1.0	48.1 $\pm$ 0.5	<u>77.4$\pm$0.2</u>	77.3	78.9$\pm$0.1
ACM	/	74.0 $\pm$ 0.1	92.1 $\pm$ 0.7	46.5 $\pm$ 1.6	50.8 $\pm$ 0.2	50.4 $\pm$ 1.2	80.6 $\pm$ 0.8	89.6	98.9$\pm$0.1
Flickr	/	74.3 $\pm$ 0.1	78.0 $\pm$ 5.0	47.9 $\pm$ 1.9	50.1 $\pm$ 0.9	49.5 $\pm$ 0.9	78.5$\pm$0.2	/	76.8 $\pm$ 0.3
Average	65.3*	64.5	67.4	49.5	54.7*	49.6	<u>68.5</u>	76.2*	81.8

TABLE III: AUROC (%) of every applicable library detector on every benchmark (mean over 5 seeds; “-” = precondition not met, so the detector does not run), and the composed result. Best single detector per dataset in bold. T = type as in Table I (C closed-form, F fast, D deep).

Detector	T	Enron	Reddit	Books	Disney	Weibo	Cora	Amazon	BlogCatalog	ACM	Flickr
reconstruction	D	29.5	50.6	48.2	42.2	86.2	71.9	74.2	74.6	74.8	74.4
contrastive	D	45.1	50.8	53.5	47.1	40.1	38.1	38.0	43.6	39.4	39.9
adversarial	D	47.8	54.7	50.0	-	64.0	36.2	32.5	43.0	26.9	34.4
one-class	D	61.8	49.5	49.4	61.0	75.4	50.0	50.0	50.0	50.0	50.0
subgraph contrast	D	-	-	-	-	90.1	87.5	93.2	77.8	80.9	78.4
local affinity	F	38.1	58.3	48.7	43.1	25.7	86.5	79.0	43.3	79.1	37.2
density	F	29.7	57.2	35.4	52.4	57.9	69.9	55.6	68.6	72.0	69.5
community	F	38.9	57.9	31.8	56.9	60.1	73.9	74.2	74.7	75.4	75.2
homophily violation	F	46.4	59.1	36.5	48.7	83.0	63.7	73.3	74.2	78.3	72.3
spectral	F	26.0	59.1	56.5	55.8	70.8	39.5	74.8	72.3	76.0	70.6
attr-struct mismatch	F	50.2	49.6	40.8	36.7	38.8	42.5	49.3	44.8	61.0	47.2
conformity	C	83.7	-	-	-	-	-	-	-	-	-
linear residual	C	-	-	-	-	94.9	-	-	-	-	-
peripheral mismatch	C	-	-	-	86.4	-	-	-	-	-	-
attribute inflation	C	-	-	72.1	-	-	-	-	-	-	-
clique density	C	-	-	-	-	-	76.2	-	60.6	73.6	59.9
Fused (gate)		83.7	59.8	72.1	86.4	94.9	93.6	73.2	78.9	98.9	76.8

detector run outside its regime is not just weak, it is anti-signal, which is why preconditions and selection exist at all. Second, comparing the fused row with the per-column best shows that the label-free composition tracks the best available detector remarkably well: it matches or exceeds the best single detector on 8 of the 10 benchmarks: four exact matches where one trusted detector carries the graph, and four wins where composition adds something, most clearly where two anomaly populations coexist and the union retrieves both (ACM 98.9 vs. 80.9; Cora 93.6 vs. 87.5). The exceptions are Flickr (1.6 below) and Amazon (73.2 vs. 93.2), where the pool dilutes one strong view. Third, because fusion is the last step, all of these rankings are part of ARCADE’s output: a practitioner who can verify a handful of top-ranked candidates (an expert inspecting each list) can make the final fused-versus-single choice per graph.

B. The analyst seat: 10 models, same evidence

Table IV is the central result of the analyst study. The study uses the 8 benchmarks that have a single mechanism-critical paradigm; Reddit (no single paradigm separates its anomalies, Table III) and Cora (its answer is a two-population union, not one key view) do not admit the Key metric. With the full evidence interface, qwen3:8b, a free 5 GB local model, identifies the mechanism-critical paradigm on all 8 benchmarks, averages 84.0, and exceeds the run-everything

TABLE IV: Analyst study on 8 benchmarks with the full evidence interface. Key = benchmarks (of 8) where the model selects the mechanism-critical paradigm in ≥ 2 of 3 repeats. Valid = parseable decisions. Time = median per decision; for local models this is on-device inference, for hosted models (Claude, DeepSeek) it is an API round-trip including network and provider serving, so the two groups are not comparable with each other. Lower block: non-LLM selection policies replayed through the same gate on the same cached scores. Random is the exact expectation over all 5,240 possible 1–4 selections across the 8 benchmarks; the fixed subset is chosen in hindsight with labels, an oracle bound for any static rule.

Analyst	AUROC	Key	Valid	Med. latency
qwen3:8b	84.0	8/8	24/24	225 s
gemma4:12b	79.7	7/8	24/24	192 s
claude-haiku-4.5	78.8	7/8	24/24	55 s
claude-opus-4.8	75.2	6/8	24/24	61 s
deepseek-v4-flash	74.2	4/8	24/24	310 s
phi4:14b	69.8	5/8	24/24	44 s
claude-sonnet-5	69.7	1/8	23/24	85 s
gemma4:e2b	68.8	4/8	24/24	30 s
qwen3:4b	67.3	4/8	23/24	238 s
llama3.1:8b	65.8	5/8	24/24	17 s
run everything (system mode)	83.1	-	-	-
best fixed 3-subset (labels)	80.1	-	-	-
closed-form triggers only	72.7	-	-	-
random 1–4 selection	62.7	-	-	-
tail-separation top-3	62.5	-	-	-

system (83.1) while executing 1–4 detectors instead of up to 12. The analyst seat is therefore not a luxury: it delivers brute-force accuracy at a fraction of the detector budget, and it is the only label-free mode that recovers Amazon (86.0 vs. 73.2), where pooling dilutes the dominant view. Four further observations.

No simpler selection policy fills the seat. We replayed every

alternative policy through the same gate on the same cached scores (Table IV, lower block). Enumerating all 5,240 possible 1–4 selections gives the exact expectation of a random analyst: 62.7. The obvious label-free heuristic, ranking detectors by the tail separation of their score distributions, scores 62.5, below random; this is the same false signal the evidence ablation exposes (Section VI-C), measured a second way, and here it even sees the full distributions where the analyst sees only quantiles. Selecting the triggered closed-form detectors alone reaches 72.7. Even the best fixed 3-subset, chosen in hindsight with full label access (an oracle bound for any static rule), averages 80.1. On 5 of the 8 benchmarks the analyst’s selection attains the maximum AUROC achievable by any subset, so no policy could do better there. A random draw ties the analyst mainly where a fired closed-form trigger lets the gate rescue any selection containing it; where no trigger fires the probability falls to 8% (Amazon). What the seat supplies is per-graph adaptation, and nothing simpler we tested supplies it.

Neither scale nor decoding mode decides. Claude Opus 4.8, the strongest general-purpose model in this comparison, ties or beats the leaders on 6 of 8 datasets, including a three-times-identical composition on ACM at the system ceiling (98.9), but loses the average on exactly two graphs: Enron (37.3 vs. 83.7) and Books (46.4 vs. 72.1). These are the two datasets whose correct answer contradicts standard anomaly priors (on Enron the anomalies are the most *conforming* nodes). The stored rationales show the difference directly. Opus begins: “anomalies express as high-frequency, neighbor-mismatched nodes *rather than feature-space outliers*”: it dismisses the correct direction in its first sentence and then reasons fluently from distribution shapes. qwen3:8b writes: conformity “identifies inliers in feature space, *leveraging high duplication rate*”: it derives the counterintuitive inference from the grounded evidence. Explicit reasoning modes are not the ingredient: we re-ran the five mechanism benchmarks (Enron, Books, Disney, Weibo, Amazon) toggling native thinking for the local models and raising the reasoning effort of the hosted ones, and the effect is model-specific with no systematic gain in either direction (disabling thinking moves gemma4:12b from 12 to 15 correct mechanism picks and qwen3:8b from 15 to 12; raised effort changes the three Claude models by -2 , $+1$, and -3). What tracks success across all conditions is whether the stated rationale grounds the decision in the given evidence, whatever the decoding mode.

Reliability is the right metric. Under weaker evidence (Section VI-C) each small model catches a different subset of mechanisms and none is reliable across regimes; the full interface is what turns scattered hits into 7–8 of 8 for the best models. Aggregation is no substitute: the median analyst averages 74.0, and a majority vote over all stored selections, replayed through the same gate, 73.2, because the majority misses the counterintuitive mechanisms and consensus inherits the error.

Cost. A local analyst decision takes 17–238 s on one consumer GPU; hosted decisions take 55–310 s of API round-trip

TABLE V: Evidence ablation on the benchmark whose mechanism (conformity: the anomalies are the most template-like nodes) contradicts standard priors. Cells: analysts (of the 8 evaluated under all conditions) that select the mechanism in ≥ 2 of 3 repeats. Selecting it yields AUROC 83.7; missing it yields 20–46.

Evidence condition	Analysts correct
duplicate-row rate hidden	0/8
field exposed, no grounding	4/8
field + grounding + <i>quality scores</i>	1/8
field + grounding (final interface)	3/8

(network and provider serving included, so hosted and local times are not comparable) and cost \$0.01–0.04. Each deep detector a decision avoids costs minutes of GPU time.

C. Evidence-interface ablations

Table V isolates the two halves of diagnosability. With the decisive field hidden, no analyst succeeds: the failure is uniform across model scales, so it belongs to the interface, not the models. Exposing the field flips the deliberate reasoners. Adding a per-paradigm quality score then *reverses* the gains: models rank paradigms by the number instead of reasoning about mechanisms (one stored rationale argues a paradigm is “strongly supported by a high tail separation score”), and the score is a distribution property, not detector quality; on this graph it argues for the wrong paradigms. Evidence design is not monotone: what you add matters as much as what you expose. The conditions also interact with models: two analysts that succeed without grounding lose the mechanism when grounding is added, while phi4:14b succeeds *only* with grounding. The interface therefore cannot be tuned to one model class. We keep grounding in the final interface because the full analyst study (Table IV) is run with it and shows its overall benefit; no single evidence condition is best for every model on every graph, and the table reports this honestly.

The probe shows the same effect from the other side. The deep subgraph-contrastive view is the right choice on Amazon, but with no evidence about it, only one analyst of eight ever selects it. With the 5-epoch probe in the pack, nine of ten analysts select it in 3 of 3 repeats, and the tenth selects it once. One cheap preview turns an invisible detector into a reachable one.

D. Selection protocol

The checklist protocol (a verdict per paradigm before selecting) changes weak models most: llama3.1:8b, the weakest free-form analyst, becomes correct on 4 of 5 mechanism benchmarks under the checklist, including Enron (3/3), which it never solves free-form under any evidence condition. Verification is easier than recall. The protocol is not uniformly better: phi4:14b loses one mechanism it finds free-form, and claude-haiku fails to emit valid checklist output on one dataset. The protocol, like the evidence, interacts with the model and should be chosen per deployment rather than assumed.

E. Decision consistency

Sampled repeats answer a practical question: does the analyst decide the same way twice? Mostly not. Under sampling, identical re-decisions occur in only 0–30% of cases for the local models, and the hosted models vary at their default

temperature; on one graph claude-haiku’s three repeats span AUROC 26.4 to 94.9. Consistency improves with evidence: claude-sonnet’s only three-times-identical decision occurs exactly where the decisive evidence field is present. Instability is therefore partly a property of underdetermined evidence, not only of the model; greedy decoding and the checklist protocol are the practical mitigations.

F. Gate ablations

Replacing the union rule by a weighted average for trusted structural+contextual pairs costs 12.1 AUROC on ACM (86.8 vs. 98.9) and 2.4 on Cora; the two injected populations are retrieved separately by the union and diluted by the average. Disabling dominant selection costs 4.8 on the Amazon trusted pair (88.9 vs. 93.7; this replays the two-paradigm pair alone, while the analyst study’s three-paradigm Amazon selection composes by average-of-maxima at 86.0). Removing the companion checks admits a degenerate companion (own tail separation 0.000, against 0.74–0.83 for genuine companions) into the union on Flickr, dropping affected compositions from 74.5 to 47.8. Making fusion order-invariant removes hidden variance (Reddit std 2.3→0.7 across seeds). Replaying the 255 stored selections of all 11 arms (models and baseline policies) through the final gate raises eight arm averages by 0.7 to 5.9 AUROC and moves the other three by less than 0.2.

VII. DISCUSSION AND LIMITATIONS

What the analyst contributes. On graphs where a closed-form precondition fires, the gate composes correctly whatever is selected, and the analyst adds little; on graphs where the right view is a deep detector with no trigger, the analyst is the only component that can find it, and with full evidence it does. The two deployments are complementary: the run-everything mode needs no language model and more compute; the analyst mode needs one good local model and a fraction of the detector budget.

Limitations. Amazon shows the run-everything mode diluting a dominant deep view; our label-free dominant-selection rule is deliberately conservative and does not fire in large pools. This is not a fixable fusion formula; when a pool holds many distinct mediocre views, the score distributions alone do not say whether one view should rule or the views should ensemble. Pooling indiscriminately erases the very information needed to prefer one view; recovering such cases without labels requires selecting *before* pooling, which is what the analyst mode provides; and because both the fused and the per-detector rankings are always output (Table III), an expert with a few verified anomalies can also make the call directly. On Flickr ARCADE stays 1.7 below the self-run SL-GAD, and the published social numbers of both baselines exceed what we could reproduce under their reference budgets; doubling our own evaluation budget does not change our results, so we report the self-run comparison and disclose the difference. The benchmarks are dominated by standard injections; organic-anomaly graphs are fewer and remain the harder case (Reddit 59.8). The analyst models are fixed snapshots; hosted model

behavior can change across versions, which our logged traces at least make auditable. Finally, evidence design has an adversarial flip side we do not study: an interface that can make mechanisms reachable could also be crafted to mislead.

VIII. CONCLUSION

ARCADE treats unsupervised anomaly detection on graphs as a reasoning problem: a covering library, label-free evidence, an exchangeable language-model analyst, and a label-free composition gate. Four constraints (coverage, diagnosability, complementarity, fusability) turn into concrete design rules, each validated by ablations of the final system. The system leads self-run baselines on 8 of 10 benchmarks (mean 81.8), and the analyst study yields the finding we consider most useful: the analyst seat does not require a frontier model, it requires the right evidence. Given that, an 8B local model reads every mechanism correctly and beats running everything. The evidence interface, not the model, is the part of such systems that deserves the most design attention, and it is fully measurable.

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